BAC Cram Pack - BAC STE Tutor

Q1. [maths | Limits and continuity] What is lim(x->0) sin(x) / x ?
A. 0
B. 1
C. inf
D. -1
Answer: B
This is a standard limit. As x approaches 0, sin(x) behaves like x, so the ratio tends to 1. It is the key limit behind the derivative of sin.

Q2. [maths | Derivatives] If f(x) = x? - 3x, what is f?(1)?
A. 0
B. 3
C. -2
D. 6
Answer: A
Differentiate term by term: f?(x) = 3x? - 3. At x = 1, f?(1) = 3 - 3 = 0. The tangent is horizontal there.

Q3. [maths | Derivatives] What is the derivative of ln(x? + 1)?
A. 1/(x? + 1)
B. 2x/(x? + 1)
C. 2x ln(x? + 1)
D. x/(x? + 1)
Answer: B
Use the chain rule: d/dx ln(u) = u?/u with u = x? + 1. Since u? = 2x, the derivative is 2x/(x? + 1).

Q4. [maths | Integrals] Evaluate ??? 2x dx.
A. 1
B. 2
C. 1/2
D. 0
Answer: A
An antiderivative of 2x is x?. Evaluate from 0 to 1: 1? - 0? = 1. It is also the area of a triangle with base 1 and height 2, so 1.

Q5. [maths | Sequences] An arithmetic sequence has u? = 3 + 2n. What is u???
A. 20
B. 23
C. 25
D. 30
Answer: B
Substitute n = 10: u?? = 3 + 2x10 = 23. In an arithmetic sequence each term grows by the common difference 2.

Q6. [maths | Complex numbers] Compute (2 + 3i)(1 - i).
A. 5 + i

C. -1 + 5i
D. 2 - 3i
Answer: A
Expand: 2 - 2i + 3i - 3i?. Since i? = -1, -3i? = +3, so the result is (2 + 3) + (-2 + 3)i = 5 + i.

Q7. [maths | Probability] Two fair dice are rolled. What is the probability that the sum is 7?
A. 1/12
B. 1/6
C. 1/9
D. 5/36
Answer: B
There are 36 equally likely outcomes. The pairs summing to 7 are (1,6), (2,5), (3,4), (4,3), (5,2) and (6,1): 6 outcomes. So P = 6/36 = 1/6.

Q8. [maths | Exponentials and logarithms] Solve 2? = 8.
A. 2
B. 3
C. 4
D. 8
Answer: B
Write 8 as a power of 2: 8 = 2?. Since the bases are equal, the exponents are equal, so x = 3.

Q9. [maths | Exponentials and logarithms] What is log?(32)?
A. 4
B. 5
C. 6
D. 16
Answer: B
log?(32) asks: 2 raised to what power gives 32? Since 2? = 32, the answer is 5.

Q10. [maths | Differential equations] Solve y? = 2y with y(0) = 3.
A. y = 3e??
B. y = e?? + 2
C. y = 3e???
D. y = 6x + 3
Answer: A
The general solution of y? = ky is y = Ce^{kx}. Here k = 2, so y = Ce??. Using y(0) = 3 gives C = 3, hence y = 3e??.

Q11. [maths | Limits and continuity] The function f(x) = (x? - 1)/(x - 1) is defined for x ? 1. Which value of f(1) makes it continuo
A. 0
B. 1
C. 2
D. It cannot be made continuous
Answer: C

= 2. Setting f(1) = 2 removes the hole.

Q12. [maths | Geometry in space] What is the volume of a sphere of radius 3?
A. 12?
B. 36?
C. 27?
D. 9?
Answer: B
The volume of a sphere is (4/3)?r?. With r = 3: (4/3)? x 27 = 36?.

Q13. [english | Conditionals] If I ___ harder, I would have passed the exam.
A. studied
B. had studied
C. study
D. would study
Answer: B
This is a third conditional: past unreal situation. The pattern is If + past perfect, would have + past participle. So: If I had studied harder, I would have passed.

Q14. [english | Tenses] She ___ in London since 2019.
A. lives
B. lived
C. has lived
D. is living
Answer: C
Since 2019 marks a point in the past that continues to now, so use the present perfect: has lived. The present perfect connects past to present.

Q15. [english | Passive voice] The report ___ by the manager yesterday.
A. wrote
B. was written
C. has written
D. is writing
Answer: B
The report receives the action, so use the passive. Yesterday is past, so the past simple passive is was + past participle: was written.

Q16. [english | Reported speech] He said, "I am tired." -> He said he ___ tired.
A. is
B. was
C. has been
D. will be
Answer: B
When the reporting verb is in the past (said), the tense usually shifts back one step: am becomes was. So: He said he was tired.

Q17. [english | Modals] You ___ smoke here. It is forbidden.

B. don’t have to
C. needn’t
D. shouldn’t have
Answer: A
Mustn’t expresses prohibition: it is not allowed. Don’t have to and needn’t mean it is not necessary, which is different.

Q18. [english | Relative clauses] The book ___ I bought yesterday is excellent.
A. who
B. which
C. whose
D. where
Answer: B
The antecedent is a thing (the book), so use which (or that). Who is for people, whose shows possession, and where is for places.

Q19. [english | Vocabulary] Choose the closest synonym of "achieve".
A. accomplish
B. abandon
C. avoid
D. admit
Answer: A
To achieve means to succeed in reaching a goal. Accomplish has the same meaning. Abandon and avoid are close to the opposite.

Q20. [english | Collocations] We need to ___ a decision before Friday.
A. do
B. make
C. take
D. give
Answer: B
The standard collocation is make a decision. English uses make for decisions, plans and choices, while do is used for tasks and activities.

Q21. [english | Prepositions] She is very interested ___ astronomy.
A. on
B. in
C. at
D. for
Answer: B
The adjective interested is followed by in: interested in something. Learn adjective + preposition pairs as fixed chunks.

Q22. [english | Gerunds and infinitives] I enjoy ___ novels in the evening.
A. to read
B. reading
C. read

Answer: B
The verb enjoy is followed by a gerund (-ing form): enjoy reading. Verbs like want and decide take the infinitive instead.

Q23. [english | Tenses] By the time we arrived, the film ___.
A. started
B. has started
C. had started
D. was starting
Answer: C
One past action happened before another past action. Use the past perfect for the earlier one: the film had started before we arrived.

Q24. [english | Word formation] What is the adjective formed from "environment"?
A. environmental
B. environmentally
C. environmentalist
D. environ
Answer: A
The adjective is environmental (environment + -al), as in environmental issues. Environmentally is the adverb, and environmentalist is a person.

Q25. [maths | Limits and continuity] What is lim(x->2) (x? - 4)/(x - 2)?
A. 0
B. 2
C. 4
D. The limit does not exist
Answer: C
Factor the numerator: x? - 4 = (x - 2)(x + 2). For x ? 2 the expression equals x + 2, so the limit is 2 + 2 = 4.

Q26. [maths | Derivatives] If f(x) = x?e?, what is f?(x)?
A. 2xe?
B. x?e?
C. e?(x? + 2x)
D. e?(x? - 2x)
Answer: C
Use the product rule: (uv)? = u?v + uv? with u = x? and v = e?. Then f? = 2xe? + x?e? = e?(x? + 2x).

Q27. [maths | Derivatives] What is the derivative of cos(2x)?
A. sin(2x)
B. -sin(2x)
C. -2 sin(2x)
D. 2 cos(2x)
Answer: C
Use the chain rule: d/dx cos(u) = -u? sin(u). With u = 2x, u? = 2, so the derivative is

BAC Cram Pack - BAC STE Tutor

Q28. [maths | Integrals] What is ? (1/x) dx for x > 0?
A. ln(x) + C
B. 1/x? + C
C. x ln(x) + C
D. -1/x? + C
Answer: A
The function whose derivative is 1/x is ln(x). For x < 0 you write ln|x|. So the antiderivative is ln(x) + C on x > 0.

Q29. [maths | Integrals] Evaluate ??^(?/2) cos(x) dx.
A. 0
B. 1
C. ?/2
D. 2
Answer: B
An antiderivative of cos(x) is sin(x). Evaluate from 0 to ?/2: sin(?/2) - sin(0) = 1 - 0 = 1.

Q30. [maths | Sequences] A geometric sequence has u? = 2 x 3?. What is u??
A. 18
B. 24
C. 54
D. 72
Answer: C
Substitute n = 3: u? = 2 x 3? = 2 x 27 = 54. Each term is three times the previous one.

Q31. [maths | Complex numbers] What is the modulus of z = 3 + 4i?
A. 3
B. 4
C. 5
D. 7
Answer: C
The modulus is |z| = sqrt(a? + b?) = sqrt(3? + 4?) = sqrt(9 + 16) = sqrt25 = 5. It is the distance from the origin.

Q32. [maths | Probability] A and B are independent with P(A) = 0.5 and P(B) = 0.4. What is P(A ? B)?
A. 0.1
B. 0.2
C. 0.4
D. 0.9
Answer: B
For independent events, P(A ? B) = P(A) x P(B) = 0.5 x 0.4 = 0.2. Independence lets you multiply the probabilities.

Q33. [english | Articles] She is ___ engineer at a large company.
A. a

C. the
D. no article
Answer: B
Engineer starts with a vowel sound, so use an. We say an engineer, just as we say an apple and an hour.

Q34. [english | Comparatives] This exam is ___ than the last one.
A. more hard
B. harder
C. hardest
D. most hard
Answer: B
Short adjectives form the comparative with -er: hard -> harder. Use more only with longer adjectives, such as more difficult.

Q35. [english | Phrasal verbs] I need to ___ up on my vocabulary before the exam.
A. brush
B. look
C. take
D. put
Answer: A
To brush up on something means to revise or improve a skill you already have. Look up means to search for information.

Q36. [english | Linking words] He studied hard; ___, he passed the exam.
A. however
B. therefore
C. although
D. unless
Answer: B
The second clause is a result of the first, so use therefore. However introduces a contrast, and although introduces a concession.

Q37. [english | Word formation] What is the noun formed from the verb "decide"?
A. deciding
B. decisive
C. decision
D. decidedly
Answer: C
The noun is decision. Decisive is an adjective, decidedly is an adverb, and deciding is a participle or gerund.

Q38. [english | Tenses] I ___ my homework when the phone rang.
A. did
B. was doing
C. have done
D. had done

The past continuous (was doing) describes an action in progress that was interrupted by a shorter past action (rang).

Q39. [english | Conditionals] If it rains tomorrow, we ___ at home.
A. stayed
B. would stay
C. will stay
D. had stayed
Answer: C
This is a first conditional: a real future possibility. The pattern is If + present simple, will + base verb.

Q40. [english | Passive voice] The results ___ tomorrow morning.
A. announce
B. will announce
C. will be announced
D. are announcing
Answer: C
The results receive the action, so use the passive. For a future action: will be + past participle = will be announced.

Q41. [maths | Derivatives] What is the derivative of cos(x)?
A. sin(x)
B. -sin(x)
C. -cos(x)
D. 1 / sin(x)
Answer: B
The derivative of cos(x) is -sin(x). The minus sign matters: cos starts at 1 and decreases, so its slope at 0 is 0 and becomes negative just after.

Q42. [maths | Derivatives] What is the tangent line to f(x) = x? at x = 3?
A. y = 6x - 9
B. y = 6x + 9
C. y = 9x - 6
D. y = 3x
Answer: A
f(3) = 9 and f?(x) = 2x, so f?(3) = 6. Point-slope form: y - 9 = 6(x - 3), which simplifies to y = 6x - 9.

Q43. [maths | Integrals] Evaluate ??^? sin(x) dx.
A. 0
B. 1
C. 2
D. ?
Answer: C
An antiderivative of sin(x) is -cos(x). Evaluate from 0 to ?: -cos(?) - (-cos(0)) = 1 + 1 = 2.

BAC Cram Pack - BAC STE Tutor

Q44. [maths | Sequences] An arithmetic sequence has first term 1 and common difference 2. What is the sum of the first 10 te

A. 90
B. 100
C. 110
D. 120

Answer: B

Use S? = n/2 x (2u? + (n - 1)d). With n = 10: S?? = 5 x (2 + 18) = 5 x 20 = 100. The terms are the first ten odd numbers, which average 10.

Q45. [maths | Complex numbers] What is i? ?

A. 1
B. -1
C. i
D. -i

Answer: B

By definition i is a number whose square is -1, so i? = -1. Everything else follows: i? = -i and i? = 1, then the pattern repeats.

Q46. [maths | Probability] A fair die is rolled once. What is the probability of an even number?

A. 1/6
B. 1/3
C. 1/2
D. 2/3

Answer: C

The even results are 2, 4 and 6, so there are 3 favourable outcomes out of 6 equally likely ones. P = 3/6 = 1/2.

Q47. [maths | Exponentials and logarithms] What is ln(e?) ?

A. 1
B. 3
C. e?
D. 3e

Answer: B

The natural logarithm is the inverse of the exponential, so ln(e?) = x for every real x. Here x = 3.

Q48. [maths | Geometry in space] What is the distance between A(1, 2, 3) and B(1, 2, 7)?

A. 3
B. 4
C. 5
D. sqrt7

Answer: B

Use the distance formula in space: sqrt((?x)? + (?y)? + (?z)?). Only z changes, by 4, so the distance is sqrt16 = 4.

Q49. [maths | Limits and continuity] What is lim(x->0) (sqrt(x + 1) - 1) / x ?

A. 0

C. 1
D. The limit does not exist
Answer: B
Substitution gives 0/0, so multiply by the conjugate: ((sqrt(x+1) - 1)(sqrt(x+1) + 1)) / (x(sqrt(x+1) + 1)) = x / (x(sqrt(x+1) + 1)). Cancel x; as x -> 0 the denominator tends to 2, so the limit is 1/2.

Q50. [maths | Limits and continuity] Which of these functions is continuous at x = 0 ?
A. f(x) = 1/x
B. f(x) = |x|
C. f(x) = (x - 1) / |x - 1|
D. f(x) = ln(x)
Answer: B
|x| is defined at 0, its limit at 0 is 0, and the value equals the limit, so it is continuous there. 1/x and ln(x) are not defined at 0, and (x-1)/|x-1| jumps at x = 1.

Q51. [maths | Derivatives] What is the derivative of f(x) = x ln(x) ?
A. ln(x)
B. 1
C. ln(x) + 1
D. x + ln(x)
Answer: C
Use the product rule: (uv)? = u?v + uv? with u = x and v = ln(x). Then f?(x) = 1 x ln(x) + x x (1/x) = ln(x) + 1.

Q52. [maths | Integrals] What is ? e^(2x) dx ?
A. e^(2x) + C
B. (1/2)e^(2x) + C
C. 2e^(2x) + C
D. e^(2x)/x + C
Answer: B
Differentiate (1/2)e^(2x): the chain rule gives (1/2) x 2e^(2x) = e^(2x). So the antiderivative is (1/2)e^(2x) + C. Dividing by the inner coefficient 2 is the key step.

Q53. [maths | Sequences] What is lim(n->inf) (2n + 1) / (n + 3) ?
A. 0
B. 1
C. 2
D. inf
Answer: C
Divide numerator and denominator by n: (2 + 1/n) / (1 + 3/n). As n grows, 1/n and 3/n tend to 0, leaving 2/1 = 2.

Q54. [maths | Probability] A fair coin is flipped twice. What is the probability of exactly one head?
A. 1/4
B. 1/3
C. 1/2

Answer: C
The four equally likely outcomes are HH, HT, TH and TT. Exactly one head happens in HT and TH, so P = 2/4 = 1/2.

Q55. [english | Vocabulary] Choose the closest synonym of "reliable".
A. dependable
B. flexible
C. ambitious
D. curious
Answer: A
A reliable person or thing can be trusted to do what is expected, which is exactly what "dependable" means. "Flexible" is about adapting, not about trust.

Q56. [english | Collocations] She has a strong ___ of humour.
A. feeling
B. sense
C. taste
D. mind
Answer: B
The standard collocation is "a sense of humour". English pairs "sense" with humour, direction, responsibility and relief, so it has to be memorised as a chunk.

Q57. [english | Prepositions] It depends ___ the weather.
A. of
B. on
C. at
D. for
Answer: B
The verb "depend" is followed by "on" (or "upon"): depend on something. The same preposition appears in "it depends on you".

Q58. [english | Word formation] What is the noun formed from the adjective "strong"?
A. strongness
B. strength
C. strongth
D. strengthen
Answer: B
The noun is "strength". "Strengthen" is the verb, and "strongness" is not a word. This is an irregular formation worth memorising: strong -> strength -> strengthen.

Q59. [english | Phrasal verbs] The meeting was ___ until next week.
A. put off
B. put up
C. put out
D. put on
Answer: A
"Put off" means to postpone. "Put up" means to accommodate or raise, "put out" means to

BAC Cram Pack - BAC STE Tutor

Q60. [english | Articles] He is ___ honest man.
A. a
B. an
C. the
D. no article
Answer: B
Use "an" before a vowel sound, not a vowel letter. The h in "honest" is silent, so the word starts with a vowel sound: an honest man.

Q61. [english | Comparatives] The more you practise, ___ you become.
A. the best
B. the better
C. better
D. the good
Answer: B
This is the double comparative: "the more ..., the better ...". Both clauses start with "the" and use a comparative form, and the meaning is that one thing increases with the other.

Q62. [english | Tenses] I ___ my keys yesterday, so I could not open the door.
A. have lost
B. lost
C. lose
D. had lost
Answer: B
The finished time word "yesterday" points to the past simple: lost. The present perfect (have lost) is used when the time is not finished or not stated.

Limits and continuity (maths)
How to evaluate limits by substitution, factoring and conjugates, the standard limits to memorise, and what continuity means a
Key points:
- Substitute first; only simplify when you meet 0/0 or inf/inf.
- Factor for polynomials, conjugate for square roots.
- Learn the four standard limits by heart.
- Continuity needs the value and both one-sided limits to agree.

Derivatives (maths)
The derivative rules that cover the 2BAC STE exam: powers, products, quotients, the chain rule, exponentials, logs and trigon
Key points:
- Learn the table of standard derivatives.
- Product rule: u?v + uv?; quotient rule: (u?v - uv?)/v?.
- Chain rule: outside derivative times inside derivative.
- Tangent slope is f?(a); use the sign of f? for variations.

Integrals (maths)
Antiderivatives, the definite integral, area under a curve and the integration techniques tested in the 2BAC STE exam.

BAC Cram Pack - BAC STE Tutor

- Always add + C for an indefinite integral.
- Definite integral: antiderivative at the top minus at the bottom.
- Area between curves: integrate the upper minus the lower.
- Substitution undoes the chain rule; parts undoes the product rule.

Complex numbers (maths)
Algebraic and trigonometric forms, modulus and argument, conjugates, and how complex numbers solve quadratic equations.
Key points:
- Replace i? with -1 in every expansion.
- Multiply by the conjugate to divide.
- |z| = sqrt(a? + b?); the argument gives the direction.
- A negative discriminant gives two conjugate complex roots.

English tenses (english)
A compact map of the English tenses tested in the 2BAC exam, with signal words and the mistakes students make most often
Key points:
- Find the time marker before choosing a tense.
- Finished time -> past simple, not present perfect.
- when + past simple, while + past continuous.
- Past perfect marks the earlier of two past actions.

Conditionals (english)
Zero, first, second and third conditionals with the patterns, meanings and mixed forms that appear in the exam.
Key points:
- Match the time and reality of the situation, not the tense label.
- Second conditional uses past forms for an unreal present.
- Third conditional: If + had done, would have done.
- Unless means if not.

The passive voice (english)
How to form the passive in every tense, when to use it, and how to decide whether the agent belongs in the sentence.
Key points:
- Passive = be + past participle; the tense lives in be.
- Only transitive verbs (verbs with an object) can be passive.
- The by-phrase is optional; include it only when it matters.
- Do not mix active and passive forms in one clause.

Reported speech (english)
How tense, pronouns and time expressions shift when you report what someone said, plus reported questions and commands
Key points:
- Past reporting verb usually means one step back in tense.
- Change pronouns, time and place words to fit the report.
- Reported questions use statement word order and no question mark.
- Commands: told/asked + object + (not) to + verb.